

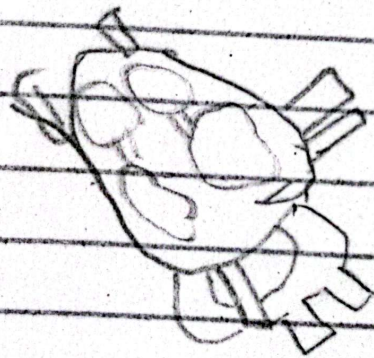
① Shannon's Theorem

$$C = B \log_2 \text{SNR}$$

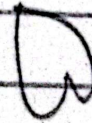


② Sampling Theorem

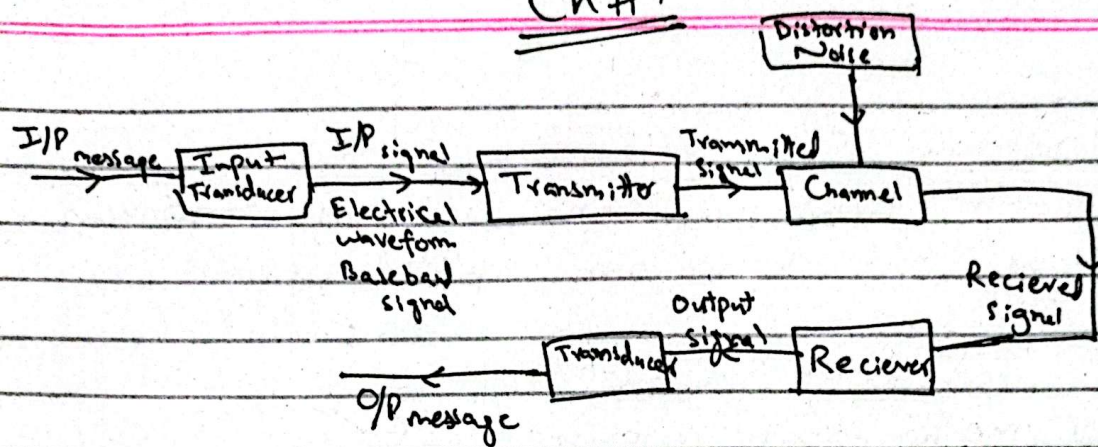
③ Nyquist Criteria for Zero ISI



↳ intersymbol interference

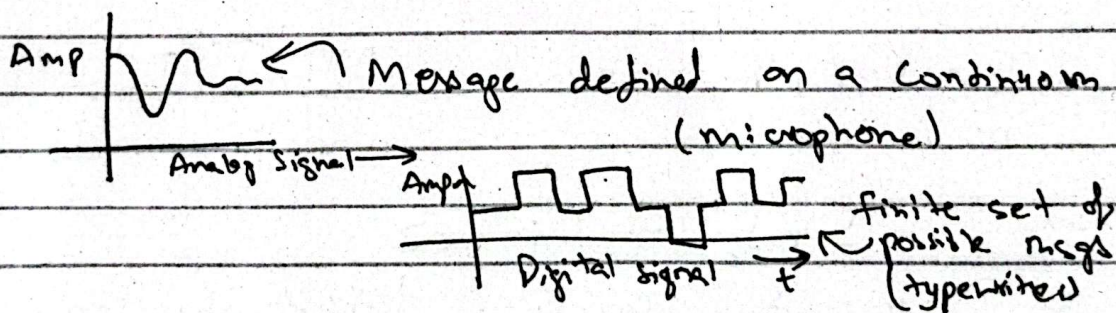


Ch#1



Communication System

SNR (Signal to noise ratio) is the ratio of signal power to noise power. Noise accumulates along the path. Noise level increases with distance from transmitter.



Analogue-to-Digital Conversion:-

- 1) Sampling
- 2) Quantization
- 3) Code Assignment to quantized levels

(From the PCM scheme - Pulse code Modulation)

The key feature of a digital communication system is that it deals with a finite set of discrete messages. In contrast to an analogue communication system in which messages are defined on a continuum.

Analog-to-Digital Conversion of Signals

Conversion of signals from analog to digital domain require

- 1) Sampling
- 2) Quantization
- 3) Code Assignment

The sampling theorem states that if the highest frequency in the signal spectrum is B (bandwidth Hz), the signal can be reconstructed from its samples taken at a rate of $2B$ samples per second.

The sample value still lies in a continuous range and can take any one of the infinite value, in the range. The problem is resolved by quantization where every one of the samples is rounded off to the nearest quantized level.

If 16 levels are assigned then we could design a code for transmission of digital

1) Analogue Signal:-

An analogue signal is any continuous time and continuous amplitude signal

$$S(t) \quad ; t \in \mathbb{R}$$

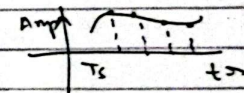
for mathematical treatment, $S \in \mathbb{C}$

2) Discrete time Signal:-

These are signals defined on a discrete time domain $\{nT_s \mid n \in \mathbb{Z}\}$ and assume continuous values on amplitude.

$T_s =$ sampling interval

n : integer



$$S(nT_s) \in \mathbb{C}; n \in \mathbb{Z}$$

Discrete time signals are derived by sampling at instants that are multiple of T_s .

3) Digital Signal:-

A digital signal is a discrete signal in both time and amplitude domains.

A digital signal satisfies

$$S(nT_s) \in A, n \in \mathbb{Z}$$

$$; A = \{Q_0, Q_1, Q_2, \dots, Q_{N-1}\}$$

with 'A' a set of N -elements.

$$\frac{1}{2\pi} \text{sinc}\left(\frac{\omega}{2}\right) \quad \text{rect}(t) \quad \frac{1}{2\pi} \text{rect}\left(\frac{t}{T}\right) \quad \frac{1}{2\pi} \text{rect}\left(\frac{t}{T}\right) \quad \frac{1}{2\pi} \text{rect}\left(\frac{t}{T}\right)$$

Shannon's Theorem

$$C = B \log_2 (1 + \text{SNR})$$

channel capacity Bandwidth Signal to noise ratio

$\Rightarrow$ Rate of Information transmission (C) $\propto B$

$\Rightarrow B$ and SNR are exchangeable

To maintain a given rate and accuracy of transmission, we trade SNR for B but not vice versa.

$$\text{SNR}_2 = \text{SNR}_1 \frac{B_1}{B_2}$$

$$\text{SNR}_2 = \text{SNR}_1 \frac{B_1}{B_2}$$

[small advantage in B buys a large advantage in SNR]

$$\text{SNR}_1 = 64, B_1 = 4 \Rightarrow B_2 = 8, \text{SNR}_2 = 8$$

Rate of Information transmission or channel or channel capacity in simplest words gives the maximum number of bits that can be transmitted per second.

With

$$P_c \rightarrow 0$$

if $N=0$, then $C \rightarrow \infty$

$$\text{SNR} = 10$$

Digit	Binary Equivalent	Pulse Code Waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

The scheme of transmitting data by digitizing, then using pulse codes to transmit the digitized data is known as pulse code modulation (PCM)

Major advantage of digital communication is the possibility of regeneration of signals

Ch # 2

Signal Energy and Energy Spectral Density

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

For complex signals: $E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$

The energy E_g of a signal $g(t)$ is defined as the area under $|g(t)|^2$.
Signal energy is also determinable from the Fourier transform $G(\omega)$ through the Parseval's theorem.
 $g(t) \rightarrow G(\omega)$

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} d\omega$$

$$E_g = \int_{-\infty}^{\infty} g(t) g^*(t) dt = \int_{-\infty}^{\infty} g(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) e^{-j\omega t} d\omega \right] dt$$

$g^*(t)$ being the conjugate of $g(t)$ can be expressed by multiplying the IFT interchanging the order of integration yields

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) \left[\int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt \right] d\omega$$

Shannon's theorem represent the upper limit on rate of communication over a channel and can not be practically achieved as the time delay and complexity $\rightarrow \infty$

$\Rightarrow$ Two primary communication source are bandwidth (B) and the transmitted power (SNR)

Significance of C:-

If the information rate $R < C$, then it is theoretically possible to achieve reliable transmission [$P_e \rightarrow 0$] through the channel by appropriate channel coding (Parity bits)
 $\hookrightarrow$ To protect data

If $R > C$, reliable transmission is not possible

E_g by Parseval's theorem

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

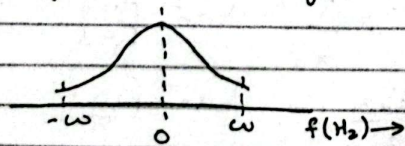
$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega$$

$$= \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-\infty}^{\infty} = \frac{1}{2a} \quad (B)$$

Hence the Parseval's theorem is verified and the signal energy E_g can be found by both time domain $g(t)$ and its Fourier transform $G(\omega)$.

Essential Bandwidth (ω) of signal:-

For most practical signals, the signal spectrum beyond ω Hz can be suppressed with little effort on the signal shape and energy.



$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) G(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

Energy Spectral density [ESD] $\Psi_g(\omega)$

$$\Psi_g(\omega) = |G(\omega)|^2$$

~~Signal Energy~~

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

$$\Psi_g(\omega) = |G(\omega)|^2$$

Q) Find the signal energy for $g(t) = e^{-at} u(t)$ $a > 0$ by the time domain specification and the equivalent signal spectrum.

$$g(t) = e^{-at} u(t)$$

$$G(\omega) = \frac{1}{j\omega + a}$$

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt = \int_0^{\infty} e^{-2at} dt = \frac{1}{2a} \rightarrow (A)$$

NORAHHH ! 20

Signal Power and Power Spectral Density (PSD)

For power signals [signals with $g(t) \rightarrow 0$ at $t \rightarrow \infty$], the signal power is meaningful as "time average of energy" averaged over an infinite time interval.

The power P_g of a signal $g(t)$ is given by

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

Signal power is thus just the time average of signal energy.

Power Spectral Density (PSD) $S_g(\omega)$:-

$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$ and consequently

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega = \frac{1}{\pi} \int_0^{\infty} S_g(\omega) d\omega$$

Example:-

Estimate the essential bandwidth ω rad/s of the signal $e^{-at} u(t)$ if the essential band is required to contain 95% of signal energy.

$$e^{-at} u(t) \Leftrightarrow \frac{1}{j\omega + a}$$

$$\text{and the ESD } \psi_g(\omega) = |G(\omega)|^2 = \frac{1}{\omega^2 + a^2}$$

$$P_g = \frac{1}{2a}$$

$$P_g = \frac{1}{2a} \times \text{Area under ESD (Parabola)}$$

$$\frac{1}{2a} = \frac{1}{2a} \times \text{Area under ESD}$$

$$0.95 \times \frac{1}{2a} = \frac{1}{2a} \int_0^{\omega} \frac{1}{\omega^2 + a^2} d\omega$$

$$\frac{0.95}{2a} = \frac{1}{2a} \left[\tan^{-1} \frac{\omega}{a} \right]_0^{\omega}$$

$$\frac{0.95}{2a} = \frac{1}{2a} \tan^{-1} \left(\frac{\omega}{a} \right)$$

$$\frac{0.95\pi}{2} = \tan^{-1} \left(\frac{\omega}{a} \right)$$

$$\omega = 12.706a \text{ rad/s}$$

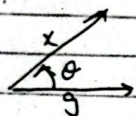
→ Spectral components of $g(t)$ from 0 to 12.706 a rad/s (2.02 a Hz) contribute 95% of signal energy.

$$f = \frac{1}{T} \quad \omega = 2\pi f$$

$$\frac{\omega}{2\pi} = \frac{1}{T} \quad \frac{\omega}{2\pi} = f$$

Correlation:-

Two vectors g and x are correlated if g has a large component along x .



smaller the θ , larger the correlation

Correlation coefficient

$$c_n = \cos \theta = \frac{g \cdot x}{|g| |x|}$$

$$-1 \leq c_n \leq 1$$

For aligned vectors ($c_n = 1$)

For vectors in opposite direction ($c_n = -1$)

For signals $g(t)$ and $x(t)$

$$c_n = \frac{1}{\sqrt{E_g E_x}} \int_{-\infty}^{\infty} g(t) x(t) dt$$

$c_n = 0$ if $g(t)$ and $x(t)$ are orthogonal.

Auto Correlation:-

The correlation of a signal with itself is called autocorrelation

$$\psi_g(t) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

Autocorrelation method for a signal $g(t)$, the energy spectral density is $|G(\omega)|^2$ can also be found by taking the Fourier

Signal Power & Power Spectral Density (PSD):-

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$$

and consequently

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

The power is $\frac{1}{2\pi}$ times the area

under the PSD.

PSD is a positive, real and even function of ω . The PSD $S_g(\omega)$ represents the power per unit bandwidth (in hertz) of the spectral components at frequency ω .

The power contributed by the spectral components within the band ω_1 to ω_2 is given by

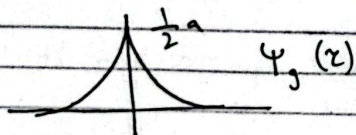
$$\Delta P_g = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega$$

Auto correlation function $\psi_g(\tau)$ is given by area under the product $g(t)g(t-\tau)$

$$\begin{aligned}\psi_g(\tau) &= \int_{-\infty}^{\infty} g(t)g(t-\tau) dt \\ &= \int_{-\infty}^{\infty} e^{-at} u(t) e^{-a(t-\tau)} u(t-\tau) dt \\ &= e^{a\tau} \int_{\tau}^{\infty} e^{-2at} dt = \frac{1}{2a} e^{-a\tau}\end{aligned}$$

For real $g(t)$, $\psi_g(\tau)$ is an even function so $-a|\tau|$

$$\psi_g(\tau) = \frac{1}{2a} e^{-a|\tau|}$$



$$\text{ESD } \psi_g(\omega) = \frac{1}{\omega^2 + a^2}$$

Transform of its auto correlation function. The reason for this alternate method is to lay a foundation for dealing with random signals.

The time auto correlation function $R_g(\tau)$ of a real power signal $g(t)$ is defined as

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t+\tau) dt$$

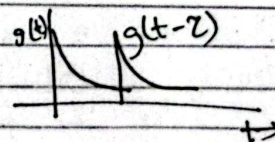
$R_g(\tau)$ is an even function of τ . This means for real $g(t)$

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g(t-\tau) dt$$

$$R_g(\tau) = R_g(-\tau)$$

Q) Find the time auto-correlation function of signal $g(t) = e^{-at} u(t)$ and from it determine the ESD of $g(t)$.

$$g(t) = e^{-at} u(t) \text{ and } g(t-\tau) = e^{-a(t-\tau)} u(t-\tau)$$



$$g(t) \Leftrightarrow G(\omega)$$

$$|G(\omega)|^2 = \Psi_g(\omega)$$

Energy Spectral Density [ESD] of I/O of LTI system

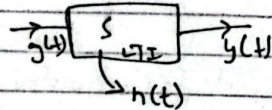
If $g(t)$ and $y(t)$ are the input and the corresponding output of a Linear time-invariant system (LTI), then

$$y(t) = g(t) * h(t)$$

$$Y(\omega) = G(\omega) H(\omega)$$

$$Y(\omega) = H(\omega) G(\omega)$$

$$|Y(\omega)|^2 = |H(\omega)|^2 |G(\omega)|^2$$



This shows that

$$\Psi_y(\omega) = |H(\omega)|^2 \Psi_g(\omega)$$

This output ESD is $|H(\omega)|^2$ times the input ESD. Since the PSD is a time avg of ESD, the input/output signal PSD of an LTI system is

$$S_y(\omega) = |H(\omega)|^2 S_g(\omega)$$

$\longleftrightarrow$

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt \quad \checkmark$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} \rightarrow \text{PSD} \quad \checkmark$$

$$P_g = \frac{1}{\pi} \int_0^\infty S_g(\omega) d\omega \quad \checkmark$$

$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega \quad \checkmark$$

Correlation:-

$$C_n = \cos \theta = \frac{g \cdot x}{|g| |x|}$$

$$C_n = \frac{1}{\sqrt{E_g E_x}} \int_{-\infty}^{\infty} g(t) x(t) dt$$

$$-1 \leq C_n \leq 1$$

$$C_n = 1 \quad ; \text{aligned}$$

$$C_n = -1 \quad ; \text{opposite}$$

$$C_n = 0 \quad ; \text{orthogonal}$$

Autocorrelation:-

$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

3.4

Signal Transmission through a Linear System

For a linear time-invariant (LTI) continuous time system, the input-output relationship is given by

$$y(t) = g(t) * h(t) \quad \text{--- (A)}$$

where $g(t) \Rightarrow$ input, $y(t) \Rightarrow$ output, $h(t) \Rightarrow$ unit-impulse response

$$g(t) \Leftrightarrow G(\omega)$$

$$y(t) \Leftrightarrow Y(\omega)$$

$$h(t) \Leftrightarrow H(\omega)$$

where $H(\omega)$ is system transfer function, then the application of the time convolution property yields

$$Y(\omega) = G(\omega) H(\omega) \quad \text{--- (B) (L.S)}$$

$H(\omega)$ thus represents here the spectral response of the system.

$$|Y(\omega)| e^{j\theta_Y(\omega)} = |G(\omega)| |H(\omega)| e^{j[\theta_G(\omega) + \theta_H(\omega)]}$$

Eq. (L.S) brings out the idea of spectral shaping or modification of the signal by the system.

Q) A noise signal $n_i(t)$ with PSD $S_{n_i}(\omega) = K$ is applied at the input of an ideal differentiator. Determine the PSD and the power of output noise signal $n_o(t)$.

Sol. $\xrightarrow{n_i(t)} \left[\frac{d}{dt} \right] \xrightarrow{n_o(t)}$

The transfer function for an ideal differentiator is

$$H(\omega) = j\omega$$

If noise at differentiator output is $n_o(t)$, then $S_{n_o}(\omega) = |H(\omega)|^2 S_{n_i}(\omega)$

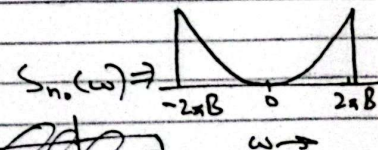
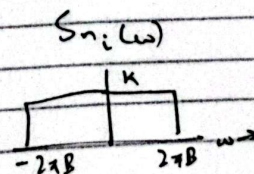
$$S_{n_o}(\omega) = |j\omega|^2 K$$

The O/P PSD is thus parabolic.

The output noise power N_o is $\frac{1}{2\pi}$ times area under the output PSD.

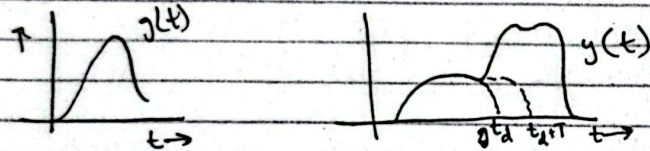
$$N_o = \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} K \omega^2 d\omega = \frac{K}{2\pi} \int_{-2\pi B}^{2\pi B} \omega^2 d\omega$$

$$= \frac{8\pi^2 B^3 K}{3}$$



⇒ A system may have flat amplitude response and get hugely distorted signal if the phase response is not linear, (t_d is not constant)

Signal Distortion over a Communication Channel



The pulse will spread if the channel characteristics is not ideal, that is

$$\theta_h(\omega) \neq -\omega t_d$$

Spreading, or dispersion, of the pulse will occur if either the amplitude response or the phase response or both are non-ideal.

Plots of $|H(\omega)|$ & $\theta_h(\omega)$ as function of ω shows how the system modifies the amplitude and the phase respectively.

Distortion Less Transmission:-

Transmission is distortion less if input and output have identical waveshapes within a multiplicative constant.

$$\text{Since } Y(\omega) = G(\omega) H(\omega)$$

$$H(\omega) = K e^{-j\omega t_d}$$

$$\rightarrow |H(\omega)| = K \rightarrow \text{constant}$$

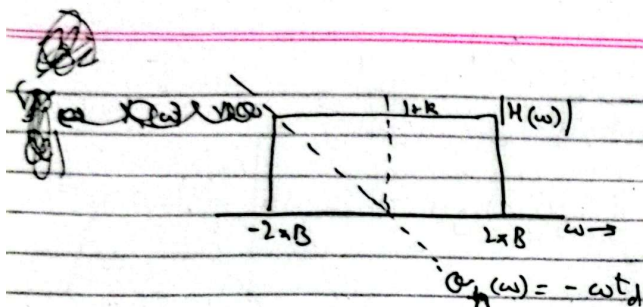
$$\theta_h(\omega) = -\omega t_d \rightarrow \text{Linear fn. of } \omega$$

A delayed output that retains the input waveform is also considered distortion less.

$$y(t) = K g(t - t_d)$$

⇒ For distortion less case, the output signal is the input signal multiplied by K and delayed by t_d .

⇒ For distortion less transmission, t_d should be constant over the band of interest



A pulse $g(t)$ band limited to B Hz is applied at the input of this filter. Find the output $y(t)$

The filter has ideal phase response and magnitude response.

$$g(t) \Leftrightarrow G(\omega)$$

$$y(t) \Leftrightarrow Y(\omega)$$

and

$$Y(\omega) = G(\omega) H(\omega)$$

$$= G(\omega) (1 + k \cos 2\omega B) e^{-j\omega t_d}$$

$$Y(\omega) = G(\omega) e^{-j\omega t_d} + k [G(\omega) \cos 2\omega B] e^{-j\omega t_d}$$

Using time shifting property,

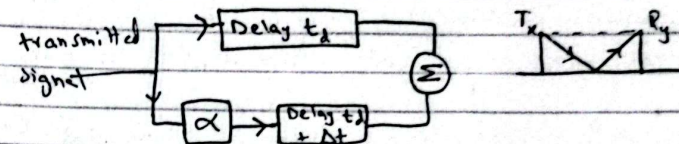
$$g(t - t_d) \Leftrightarrow G(\omega) e^{-j\omega t_d}$$

$$g(t - T) + g(t + T) \Leftrightarrow 2 G(\omega) \cos 2\omega B$$

The time domain equivalent of the equation can be expressed as

$$y(t) = g(t - t_d) + \frac{k}{2} [g(t - t_d - T) + g(t - t_d + T)]$$

Distortion Caused by multipath Effect:-



Scenarios considered has only two paths:

- 1) With a unity gain and delay t_d .
- 2) Other with gain α and delay $(t_d + \Delta t)$

Transfer function of the two paths are given by $e^{-j\omega t_d}$ and $\alpha e^{-j\omega(t_d + \Delta t)}$ respectively. Overall transfer function $H(\omega)$ is

$$H(\omega) = e^{-j\omega t_d} + \alpha e^{-j\omega(t_d + \Delta t)}$$

$$= e^{-j\omega t_d} (1 + \alpha e^{-j\omega \Delta t})$$

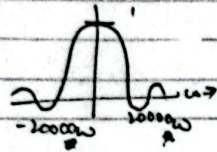
$$= e^{-j\omega t_d} (1 + \alpha \cos \omega \Delta t - j\alpha \sin \omega \Delta t)$$

Both magnitude and phase characteristics of $H(\omega)$ are periodic in ω with a period of $\frac{2\pi}{\Delta t}$.

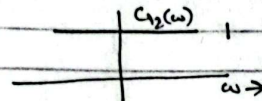
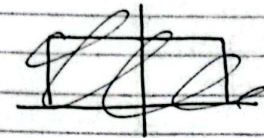
A low pass filter (shown below) has transfer function $H(\omega)$ given by

$$H(\omega) = \begin{cases} (1 + k \cos 2\omega B) e^{-j\omega t_d} & ; |\omega| < 2\pi B \\ 0 & ; |\omega| > 2\pi B \end{cases}$$

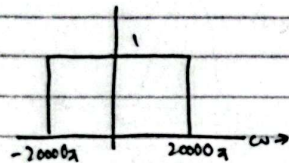
(a) $G_1(\omega) = \text{sinc}\left(\frac{\omega}{20000\pi}\right)$



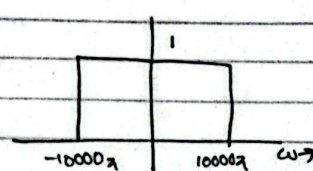
$G_2(\omega) = 1$



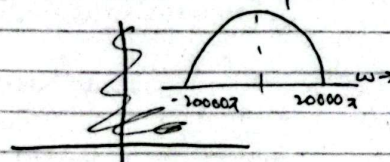
(b) $H_1(\omega)$



$H_2(\omega)$

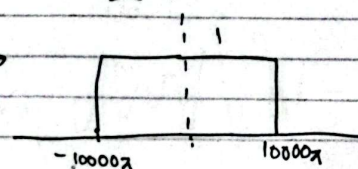


(c) $Y_1(\omega)$



$Y_1(\omega) = G_1(\omega) H_1(\omega)$

$Y_2(\omega)$



$Y_2(\omega) = G_2(\omega) H_2(\omega)$

(d) BW of $Y_1(\omega) = \frac{20000\pi}{2\pi} = 10 \text{ KHz}$

BW of $Y_2(\omega) = \frac{10000\pi}{2\pi} = 5 \text{ KHz}$

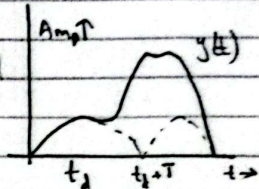
BW of $Y(\omega) = Y_1(\omega) + Y_2(\omega) = 10 + 5 = 15 \text{ KHz}$

$\frac{Y_1}{\omega} = \frac{10}{2\pi}$
 $\frac{Y_2}{\omega} = \frac{5}{2\pi}$
 $\frac{Y}{\omega} = \frac{15}{2\pi}$

The output $y(t)$ is actually

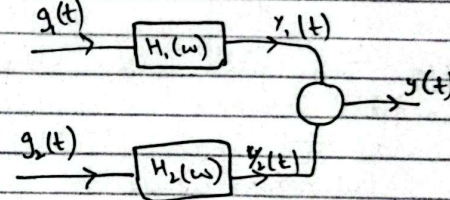
$g(t) = \frac{k}{2} [g(t-T) + g(t+T)]$ delayed

by t_d .



It consists of $g(t)$ and its echoes shifted by $\pm t_d$.

(Q)



Signals $g_1(t) = 10^4 \text{rect}(10^4 t)$ and $g_2(t) = \delta(t)$ are applied at the input of an ideal LPF.

$H_1(\omega) = \text{rect}\left(\frac{\omega}{40000\pi}\right)$ and $H_2(\omega) = \text{rect}\left(\frac{\omega}{20000\pi}\right)$

The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal

$y(t) = y_1(t) y_2(t)$

(a) Sketch $G_1(\omega)$ & $G_2(\omega)$

(b) Sketch $H_1(\omega)$ & $H_2(\omega)$

(c) Sketch $Y_1(\omega)$ & $Y_2(\omega)$

(d) Find BW of $y_1(t)$, $y_2(t)$, $y(t)$

~RAI

In AM, ~~amplitude~~ A varies in proportion to m(t)

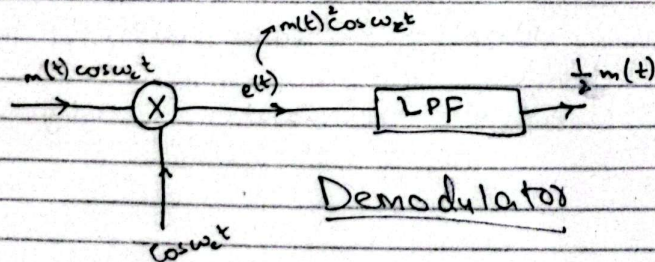
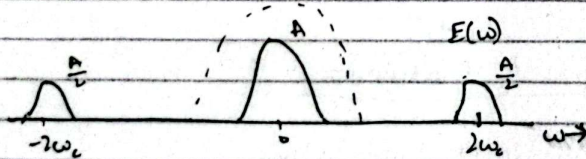
Amplitude modulation is defined as a process in which the amplitude of the carrier wave is varied about a mean value, linearly with the baseband signal m(t).

Demodulation consists of multiplication of incoming modulated signal m(t)cosω_ct by the carrier cosω_ct

$$e(t) = m(t) \cos^2 \omega_c t$$

$$= \frac{1}{2} [m(t) + m(t) \cos 2\omega_c t]$$

$$E(\omega) = \frac{1}{2} M(\omega) + \frac{1}{4} [M(\omega + 2\omega_c) + M(\omega_c - 2\omega_c)]$$



Demodulator

$$f_c = \frac{\omega_c}{2\pi}$$

Audio signal range: 0-4 kHz

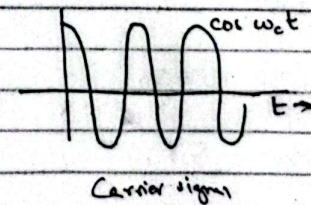
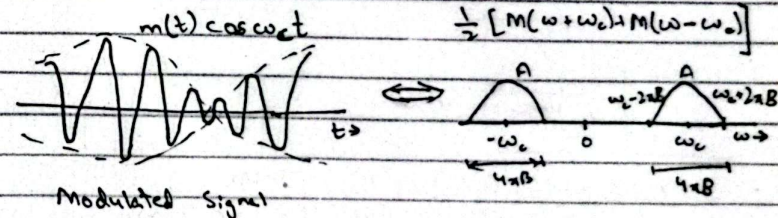
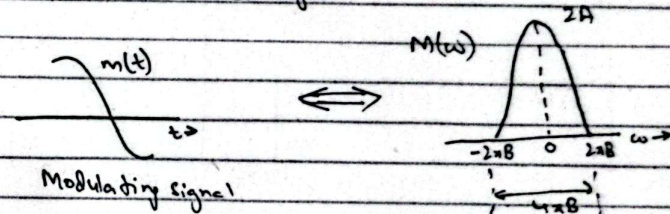
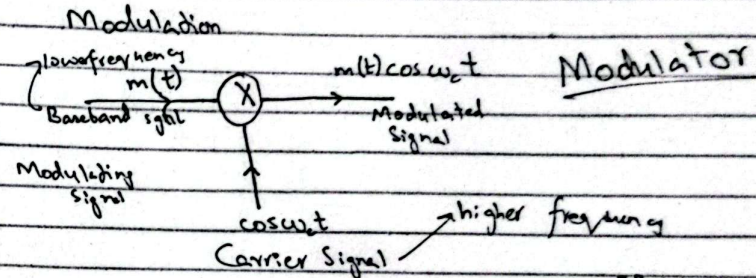
$$A \cos(\omega_c t + \theta_c)$$

Amp Freq Phase

RA

Ch#4

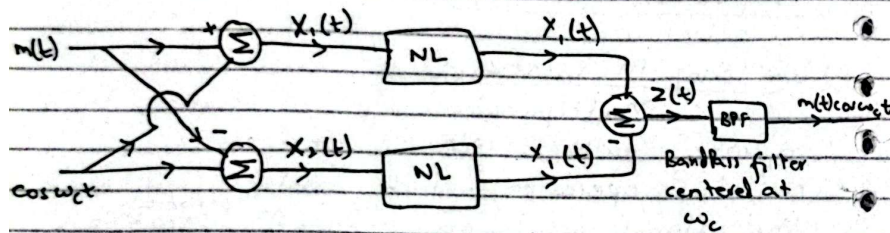
Amplitude Modulation (AM)



$$m(t) \Leftrightarrow M(\omega)$$

$$m(t) \cos \omega_c t \Leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$$

Non-Linear Modulators :- [Single Balanced Modulator]



$$y(t) = ax(t) + bx^2(t)$$

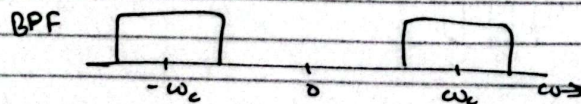
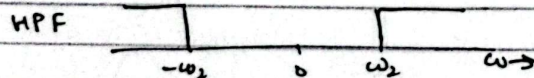
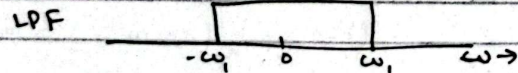
$$z(t) = y_1(t) - y_2(t) = [ax_1(t) + bx_1^2(t)] - [ax_2(t) + bx_2^2(t)]$$

$$\text{Put } x_1(t) = \cos \omega_c t + m(t)$$

$$\text{and } x_2(t) = \cos \omega_c t - m(t)$$

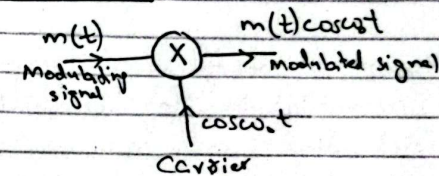
$$z(t) = 2am(t) + 4bm(t)\cos \omega_c t$$

⇒ Explanation ~~why~~ why we don't use LPF & HPF instead of BPF ⇒

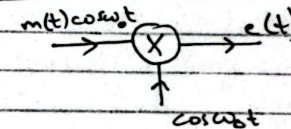


Amplitude Modulation :-

⇒ Modulator



⇒ Demodulator



Synchronous

$$e(t) = m(t) \cos \omega_c t$$

$$= \frac{1}{2} [m(t) + m(t) \cos 2\omega_c t]$$

$$= \frac{1}{2} m(t) + \frac{1}{2} m(t) \cos 2\omega_c t$$

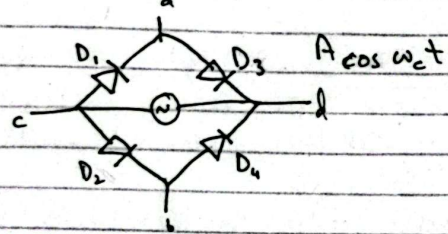
Types of Modulator :-

1) Multiplier Modulator

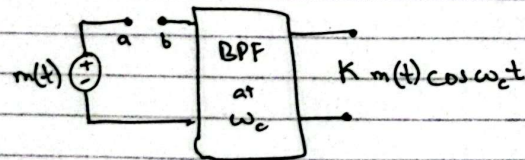
Analog multiplier with O/P proportional to the two i/p's

- Difficult to maintain linearity

The switching operation can be obtained with a diode bridge modulator (which is a double balanced modulator).

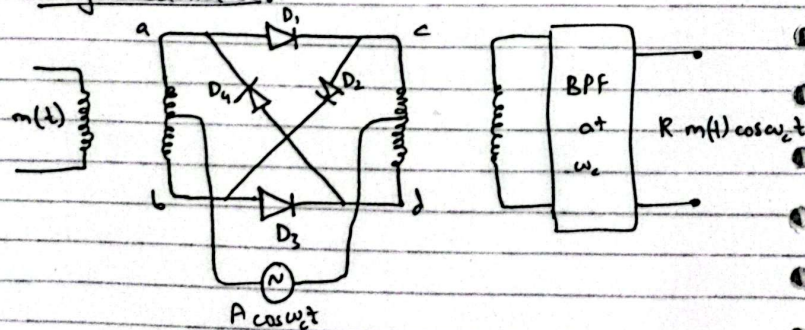


Diode bridge serves as the desired electronic switch when terminals a & b open and close periodically with carrier frequency f_c when $A \cos \omega_c t$ is applied across c & d



Switching modulator utilizing diode bridge.

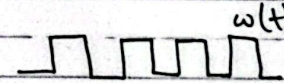
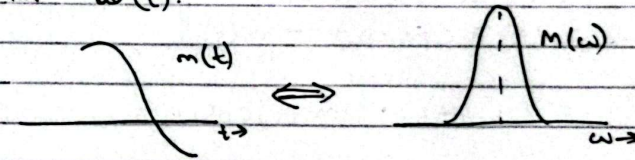
2) Ring Modulators:-



Amplitude Modulators:-

1) Switching Modulators:-

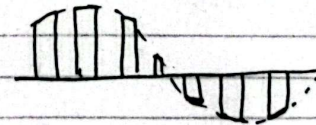
Multiplication of a signal by a square pulse train is in reality a switching operation which involves switching the signal $m(t)$ on and off ^{periodically} and can be accomplished by single switching element controlled by a pulse train $w(t)$.



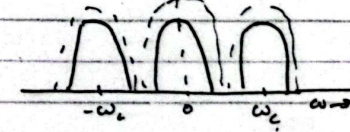
$w(t)$ is a periodic signal with Fourier series

$$w(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right)$$

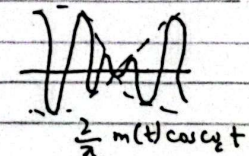
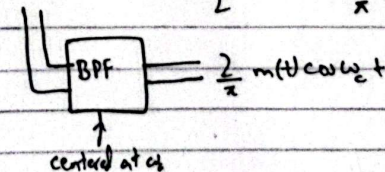
$m(t)w(t)$



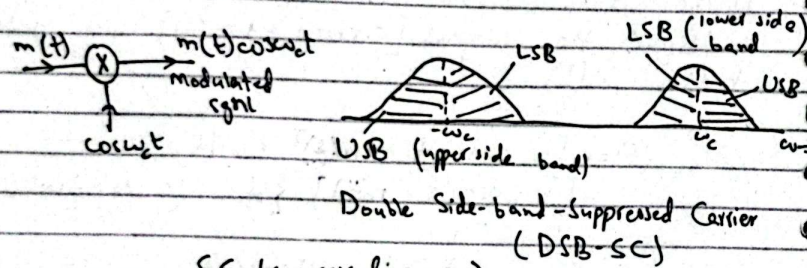
$m(\omega)$



$$m(t)w(t) = \frac{1}{2} m(t) + \frac{2}{\pi} \left(m(t) \cos \omega_c t - \frac{1}{3} m(t) \cos 3\omega_c t + \dots \right)$$



Amplitude Modulation (with carrier- ω_c)



For AM

- SC (suppressed carrier)
(costly, sophisticated receiver)
(synchronous)
- WC (point-to-multipoint system)

AM-WC
$$\phi_{AM}(t) = A \cos \omega_c t + m(t) \cos \omega_c t$$

$$= [A + m(t)] \cos \omega_c t$$

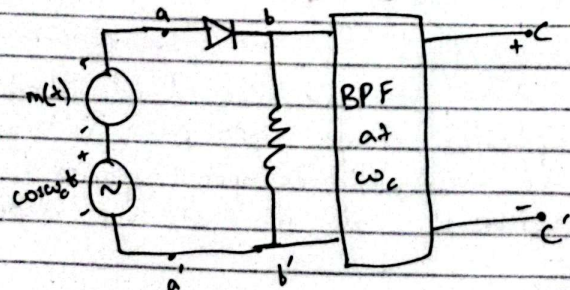
$$P_{AM}(\omega) \stackrel{\text{FT}}{=} \frac{1}{2} [M(\omega - \omega_c) + M(\omega + \omega_c)]$$

$$+ \pi A [\delta(\omega + \omega_c) + \delta(\omega - \omega_c)]$$

Two additional pulses at ω_c and $-\omega_c$

Generation of AM-WC Signal

AM signals can be generated by any DSB-SC modulator if the modulating signal is $\{A + m(t)\}$



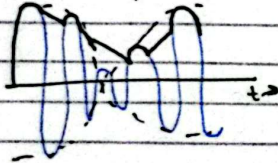
$\Rightarrow$ During +ive half cycle of the carrier; diode D_1 & D_3 conduct where as D_2 & D_4 are open. O/P proportional to $m(t)$

$\Rightarrow$ During -ive half cycle of carrier; diode D_1 & D_3 are open where as D_2 & D_4 conduct. O/P proportional to $-m(t)$

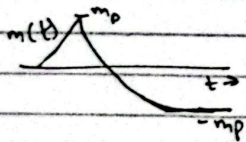
"Double side spectrum with suppressed carrier"

on the time constant RC .

The output $V_o(t)$ closely follows envelope of the input.

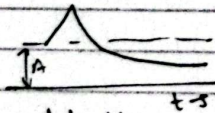


Amplitude Modulation for Broadcast System

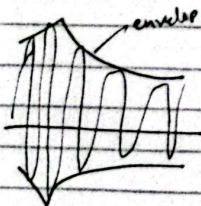


Envelope detection only if $A + m(t) > 0$

$$A + m(t) > 0$$



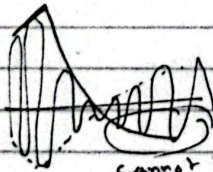
envelop detection possible



$$A + m(t) \not> 0$$



cannot be detected
envelop detection not possible
(+ive only)



cannot be detected

Input is $c \cos \omega_c t + m(t)$ with $c \gg m(t)$.

Diode acts as a switching element and multiplies the signal $[c \cos \omega_c t + m(t)]$ with pulse train $w(t)$

$$\begin{aligned} V_{bb}(t) &= [c \cos \omega_c t + m(t)] \cdot w(t) \\ &= [c \cos \omega_c t + m(t)] \left\{ \frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right) \right\} \\ &= \frac{c}{2} \cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t + \dots \end{aligned}$$

carrier

suppressed by BPF (ω_c)

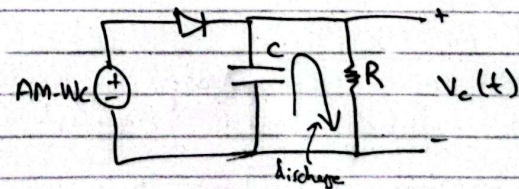
modulated signal

$$V_{cc}(t) = \frac{c}{2} \cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t$$

AM-NC

Envelope Detector:-

In an envelope detector, the output of the detector follows the envelope of the modulated signal.



→ During the +ive cycle, the diode ~~can~~ conducts and capacitor charges upto peak voltage.

→ During the -ive cycle, diode open & capacitor discharges through resistor R at a slow rate depending

→ Tone modulation (carrier is also a sine & msg is also a sine wave)

Eg 44) $\phi_{am}(t) = ?$, $\mu = 0.5$, $\mu = 1$ when $m(t) = B \cos \omega_m t$

$$\mu = \frac{m_p}{A}$$

$$\therefore m_p = B$$

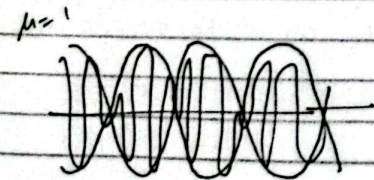
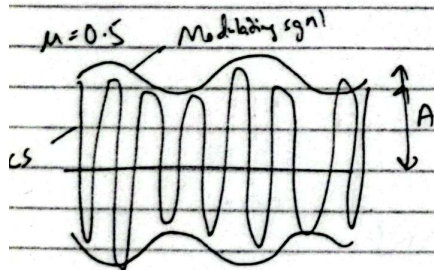
$$\mu = \frac{B}{A}$$

$$B = \mu A$$

$$\text{As } m(t) = B \cos \omega_m t$$

$$= \mu A \cos \omega_m t$$

$$\begin{aligned} \text{Hence } \phi_{am}(t) &= [A + m(t)] \cos \omega_c t \\ &= [A + \mu A \cos \omega_m t] \cos \omega_c t \\ &= A [1 + \mu \cos \omega_m t] \cos \omega_c t \end{aligned}$$



→ Condition $\Rightarrow A + m(t) \geq 0$ for all t

→ Modulation index (μ) = $\frac{m_p}{A}$
 $\xrightarrow{\text{msg peak}}$
 $\xrightarrow{\text{Ratio of Amplitude of Carrier to the peak}}$

→ $A + m(t) \geq 0$ implies that A is greater or equal to m_p ($A \geq m_p$) (case II)

Envelope detection to be possible:

$$0 \leq \mu \leq 1$$

$\Rightarrow \mu > 1$ is over modulation
 envelope det. not possible

→ Modulation Index (μ)

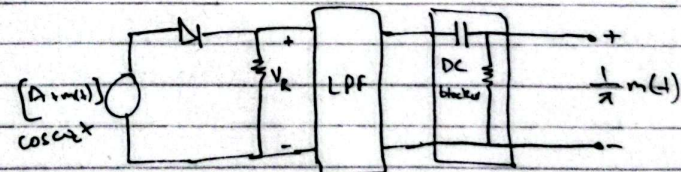
$$m(t) \geq -m_p$$

$$A - m_p \geq 0$$

$$A \geq m_p$$

Rectifier Detector (Demodulation of AM signal)

You can demodulate AM signal by using a locally carrier (coherent) but it will defeat AM (purpose) because 1



$$V_R = [A + m(t)] \cos \omega_c t \cdot \cos \omega_c t$$

$$= [(A + m(t)) \cos \omega_c t] \left[\frac{1}{2} + \frac{2}{\pi} (\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \dots) \right]$$

$\xrightarrow{\text{suppressed}}$

After LPF

$$V_{RF} = \frac{1}{\pi} (A + m(t))$$

After DC blocker (blocks A (DC shift))

$$V_{DC} = \frac{1}{\pi} m(t)$$

Ex 4.5) $\eta = ?$, a) $\mu = 0.5$ b) $\mu = 0.3$ $m = \frac{B}{A} \Rightarrow B = \mu A$

For tone modulation, $m(t) = \mu A \cos \omega_m t$
 $m^2(t) = \frac{\mu^2 A^2}{2}$

$$P_s = \frac{m^2(t)}{2} = \frac{(\mu A)^2}{2} \quad P_c = \frac{A^2}{2}$$

$$\eta = \frac{P_s}{P_c + P_s} = \frac{\frac{\mu^2 A^2}{2}}{\frac{A^2}{2} + \frac{\mu^2 A^2}{2}} = \frac{\mu^2 A^2}{A^2 + \mu^2 A^2} = \frac{\mu^2 A^2}{A^2(1 + \mu^2)} = \frac{\mu^2}{1 + \mu^2}$$

For $\mu = 0.5$

$$\eta = \frac{0.5^2}{1 + 0.5^2} \times 100\% = 11.1\%$$

For $\mu = 0.3$

$$\eta = \frac{0.3^2}{1 + 0.3^2} \times 100\% = 4.3\%$$

Sideband and Carrier Power

$$\phi_{AM}(t) = \underbrace{A \cos \omega_c t}_{\text{carrier}} + \underbrace{m(t) \cos \omega_c t}_{\text{sidebands}}$$

The carrier power P_c is mean square value of $A \cos \omega_c t$

$$P_c = \frac{A^2}{2}$$

The sideband power is power of $m(t) \cos \omega_c t$ which is

$$P_s = \frac{1}{2} m^2(t) \quad (\text{should be } +ve)$$

Power Efficiency $\eta = \frac{\text{useful power}}{\text{total power}} \times 100\%$

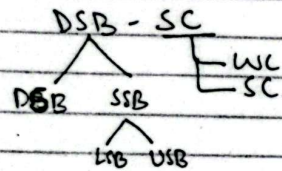
$$= \frac{P_s}{P_c + P_s} \times 100\% = \frac{\frac{1}{2} m^2(t)}{\frac{A^2}{2} + \frac{1}{2} m^2(t)} \times 100\%$$

$$\eta = \frac{m^2(t)}{A^2 + m^2(t)} \times 100\%$$

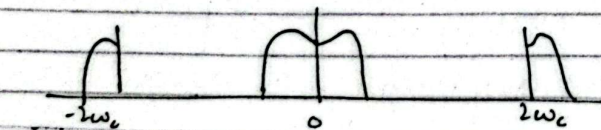
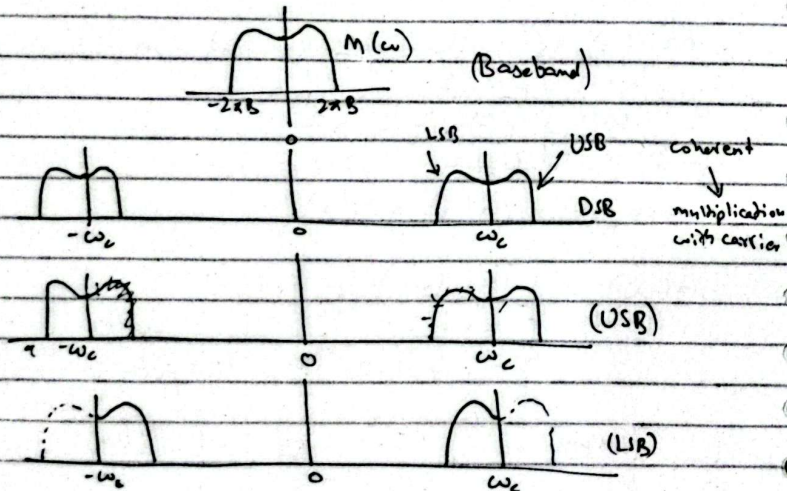
AM with carrier efficiency cannot exceed than 25%. (highly inefficient)

LC16

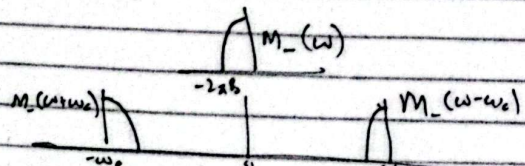
$$m(t) \cos \omega_c t$$



Single side band (reduce bandwidth)



LSB $M_-(\omega) = M(\omega) U(-\omega)$



$$\phi_{DSB}(t) = m(t) \cos \omega_c t + m_h(t) \sin \omega_c t$$

$$\phi_{SSB}(t) = m(t) \cos \omega_c t \pm m_h(t) \sin \omega_c t$$

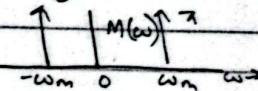
Eg 4.1) (Tone Modulation)

$$m(t) = \cos \omega_m t, \text{ DSB-SC?}$$

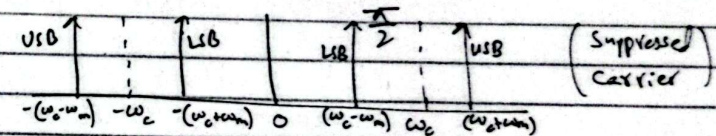
USB, LSB=?

$$m(t) = \cos \omega_m t$$

$$M(\omega) = \pi [\delta(\omega - \omega_m) + \delta(\omega + \omega_m)]$$



Spectrum consists of two impulses at $\pm \omega_c$

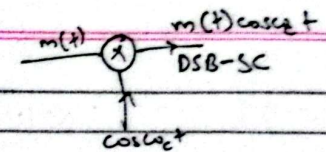


Spectrum beyond ω_c is USB and the one below is LSB.

In time domain approach we work directly with signals in time domain for baseband signal $m(t) = \cos \omega_m t$

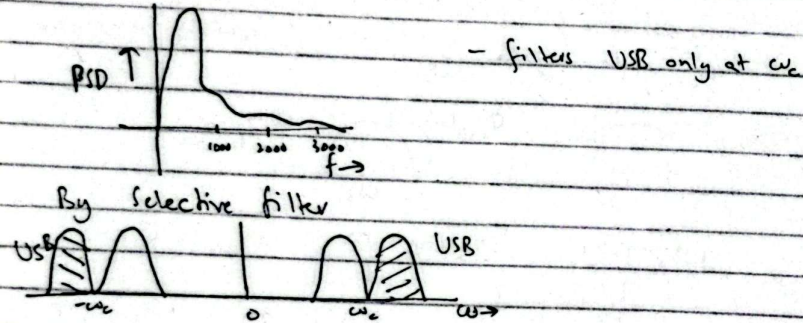
$$= \cos \omega_m t \cos \omega_c t$$

$$= \frac{1}{2} [\cos(\omega_m + \omega_c) + \cos(\omega_m - \omega_c)]$$



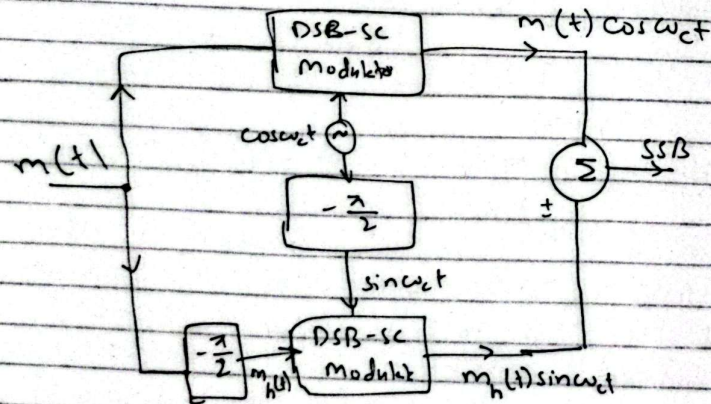
Generating SSB methods:-

1) Selective Filtering Method



2) Phase Shift Method

$$\phi_{SSB}(t) = m(t)\cos\omega_c t \pm m_n(t)\sin\omega_c t$$



3) SSB with envelop detection (slides)

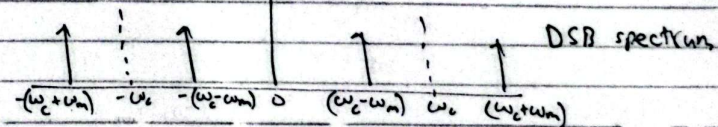
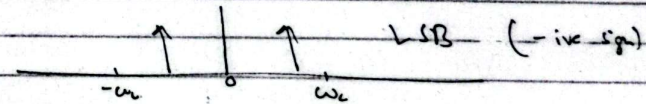
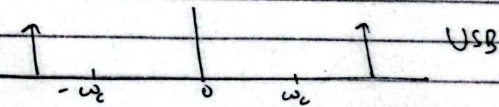
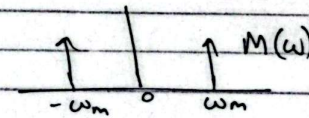
Eg 4.7) Tone Modulated Signal

$$m(t) = \cos\omega_m t$$

Find $\phi_{SSB}(t)$

$$m_n(t) = \cos(\omega_m t - \frac{\pi}{2}) = \sin\omega_m t$$

$$\begin{aligned} \phi_{SSB}(t) &= \cos\omega_m t \cos\omega_c t \pm \sin\omega_m t \sin\omega_c t \\ &= \cos(\omega_c \pm \omega_m)t \end{aligned}$$



ω_c — carrier freq.

Ch#5

Angle Modulation

"Technique of modulation where the angle of the carrier is varied with a modulating signal & $m(t)$ is known as angle modulation."

Two possibilities are phase modulation (PM) and frequency modulation (FM)

$$\phi_{PM} = A \cos(\omega_c t + K_{PM}(t))$$

$$\phi_{FM} = A \cos(\omega_c t + K_f \int_{-\infty}^t m(\alpha) d\alpha)$$

ω_i — Instantaneous frequency

Generalized Sinusoidal $A \cos(\omega_c t + \theta_i)$

Instantaneous Frequency:-

$$\phi(t) = A \cos(\omega_c t + \theta_i)$$

The instantaneous frequency ω_i at any instant t is the slope of $\theta(t)$ at t

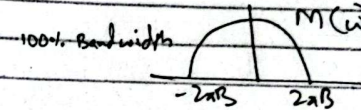
Thus for $\phi(t)$

$$\omega_i(t) = \frac{\partial \theta}{\partial t}$$

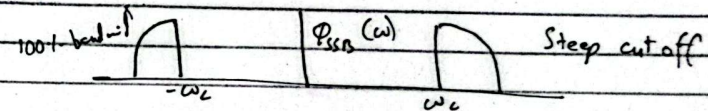
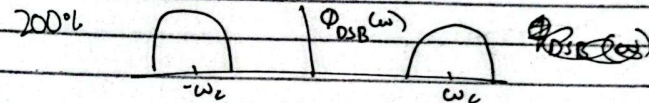
$$\theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha$$

The possibility of transmitting the information of $m(t)$ by varying angle θ of the carrier.

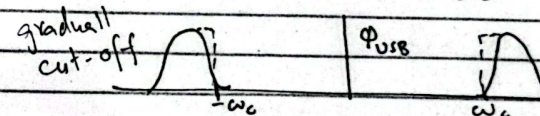
→ AM: Vestigial Sideband (VSB)



No selective filtering
Gradual filtering done



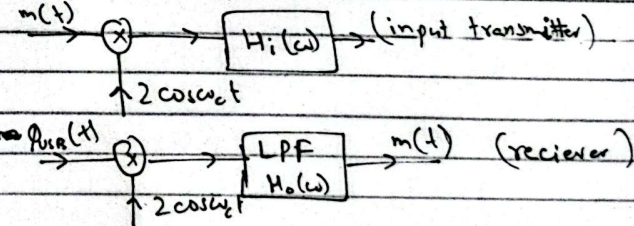
Step cut-off



125% of original
signal bandwidth

(25% extra)

VSB modulator & Demodulator:-



$H_i(\omega)$ is VSB shaping filter

• Bandwidth: 25% to 33% higher than SSB

For output side

$$e(t) = 2\phi_{um} \cdot \cos \omega_c t \Leftrightarrow [\phi_{USB}(\omega + \omega_c) + \phi_{LSB}(\omega - \omega_c)]$$

$$M(\omega) = [\phi_{USB}(\omega + \omega_c) + \phi_{LSB}(\omega - \omega_c)] H_i(\omega)$$

$$M(\omega) = M(\omega) [H_i(\omega + \omega_c) + H_i(\omega - \omega_c)] H_o(\omega)$$

$$H(\omega) = \frac{1}{H_i(\omega + \omega_c) + H_i(\omega - \omega_c)} ; |\omega| \leq 2\pi B$$

Eg 48

→ Bandwidth of Angle Modulator carrier
* generalized angle - modulated carrier

electromagnetic wave $\phi_{EM}(t) = A \cos(\omega_c t + \psi(t))$
 $= A \cos\left[\omega_c t + \int_{-\infty}^t m(\alpha) h(t-\alpha) d\alpha\right]$

If $h(t) = k_p \delta(t)$ ——— PM
 $h(t) = k_f \omega(t)$ ——— FM

Let us define $a(t) = \int_{-\infty}^t m(\alpha) d\alpha$

$\hat{\phi}_{FM}(t) = A e^{j[\omega_c t + k_f a(t)]}$
 $= A e^{j k_f a(t)} \cdot e^{j \omega_c t}$ just

$\phi_{FM}(t) = R_e \hat{\phi}_{FM}(t) = A \left[\cos \omega_c t - k_f a(t) \sin \omega_c t - \frac{k_f^2 a^2(t)}{2!} \cos \omega_c t + \frac{k_f^3 a^3(t)}{3!} \sin \omega_c t + \dots \right]$

→ Theoretical bandwidth of FM wave is infinite

NB FM $\Rightarrow$ Narrow Band FM

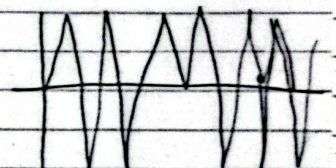
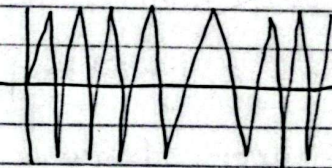
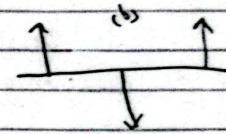
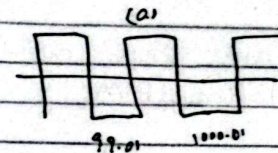
$\phi_{FM}(t) \approx A \cos(\omega_c t - k_f a(t) \sin \omega_c t)$

$\therefore |k_f a(t)| \ll 1$

$\phi_{PM}(t) = A \cos(\omega_c t + k_p m(t))$

$\phi_{FM} = A \cos\left(\omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha\right)$ increase frequency

By Eg 5.2

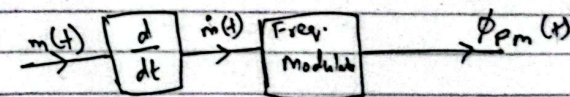
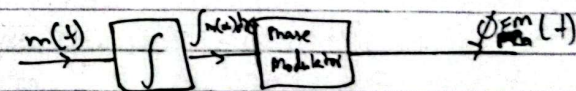


$\phi_{FM}(t)$

ϕ_{PM}

→ Instantaneous freq. varies linearly with modulating signal in FM.

Generalized Concept of Angle Modulation:-



Example) $\phi_{em}(t) = 10 \cos(\omega_c t + 5 \sin 3000t + 10 \sin 2000\pi t)$

(a) Find power of angle modulated wave
 $P = \frac{A^2}{2} = \frac{10^2}{2} = 50$

(b) Frequency deviation Δf
 $\omega_i = \frac{d}{dt} \phi(t)$

$= \omega_c + 15000 \cos 3000t + 20000\pi \cos 2000\pi t$

Carrier deviation

$15000 \cos 3000t + 20000\pi \cos 2000\pi t$

$\Delta \omega = 15000 + 20000\pi$ (add in phase)

$\Delta f = \frac{15000 + 20000\pi}{2\pi} = 12387.32 \text{ Hz}$

(c) Deviation ratio β

$\beta = \frac{\Delta f}{B}$ Bandwidth of $m(t)$

$B \Rightarrow$ highest frequency in $m(t)$

2000π in ω . $\Rightarrow \frac{2000\pi}{2\pi} B = 1000$

$\beta = \frac{12387.32}{1000} = 12.387$

(d) phase deviation $\Delta \phi$

$\phi(t) = \omega_c t + (5 \sin 3000t + 10 \sin 2000\pi t)$

$\Delta \phi = 5 + 10 = 15$ radians

(e) Bandwidth of $\phi_{em}(t)$

$B_{FM} = 2(\Delta f + B)$

$= 2(12387.32 + 1000)$

$= 26774.64 \text{ Hz}$

$B_{FM} = 2\Delta f$

$= 24774.64 \text{ Hz}$

(not correct as will over mod.)

$B_{FM} = 2B(\beta + 1)$

$= 2(1000)(12.387 + 1)$

$= 26774.64 \text{ Hz}$

Bandwidth of FM Waves

$\rightarrow$ Theoretically an FM modulated wave is not band limited.

NBFM for very small R_f
 (narrow band FM)

$\phi_{FM}(t) \approx A [\cos \omega_c t - R_f a(t) \sin \omega_c t]$
 given that $|R_f a(t)| \ll 1$

This is linear modulation. The bandwidth of $a(t)$ is B , then the BW of $\phi_{FM}(t)$ would only be $2B$.

Approximate BW for WBFM (wide band FM)

$B_{FM} = \frac{1}{2\pi} (2K_f m_p + 8\pi B)$ $\Delta f = \frac{K_f m_p}{2\pi}$

$B_{FM} = 2(\Delta f + B) \text{ --- (1)}$

For a truly wideband case, where $\Delta f \gg B$, FMBW can be approximately as

$B_{FM} \approx 2\Delta f \text{ --- (2)}$ $\Delta f \gg B$

We define the deviation ratio $\beta = \frac{\Delta f}{B}$

$B_{FM} = 2B(1 + \beta) \text{ --- (3)}$

FM - Modulation & Demodulation

Frequency modulators are $\rightarrow$ time varying $\rightarrow$ no-linear system
Oscillator (Voltage Control Oscillator VCO) are designed to produce frequency modulation.

Oscillator design $\left\{ \begin{array}{l} \text{either} \rightarrow \text{Reactance tube (older method)} \\ \text{or} \rightarrow \text{Varactor Diode (newer version)} \end{array} \right.$
 $\downarrow$
Capacitor whose capacitance changes with applied voltages